LLMs in Bioinformatics

Student pre-work

November 14-15, 2026

- Check that you can open Colab and run one Python cell.
- Local Python / VS Code setup is optional.
- Bring your laptop, charger, and questions.

Before the workshop



- Bring a laptop and charger; check your browser and network.
- Sign in to Google Colab with a Google account.
- Run the small test cell shown on the next slide.
- No paid plan, GPU, GitHub account, or API key is required for this pre-work.
- If setup fails, bring the error with private information removed.

Open Colab and test one cell



- Open colab.research.google.com and sign in.
- Create a new notebook and run the code below.
- Check the output beneath the code cell.

```
print("Hello, bioinformatics!")
```

Save your notebook. Runtime files and installed packages are temporary.

Use the slides and notebook together



- Read the slides to frame the biological question and method.
- Open the instructor-provided notebook for that activity.
- Save a copy in Drive before editing a shared workshop notebook.
- Run cells in order and inspect their outputs.
- Return to the slides to interpret the results.

The organizers will provide the final workshop notebook links.

Optional local Python setup



- Install Python 3 and VS Code.
- Install Microsoft's Python extension.
- For local notebooks, add the Jupyter extension.
- Create a .venv and select its interpreter / kernel.

Official VS Code tutorial, captured November 9, 2026.

Follow the student guide. Install lab packages only when final requirements are confirmed.

Keep research data safe

- Use only public or synthetic workshop data.
- Do not upload identifiable, clinical, confidential, unpublished, or restricted data to personal cloud notebooks or AI tools.
- Never put passwords, API keys, or access tokens into notebooks or prompts.
- Share setup errors with private details removed; a teaching assistant can help.

Optional AI access programs

GitHub Copilot Student

- Eligible enrolled students can apply through GitHub Education.
- Provide current enrollment proof, then activate Copilot after approval.
- See the guide for official eligibility and application links.

Claude for research labs

- Anthropic's scientist access program is PI-led.
- An eligible academic / nonprofit PI verifies first and may add lab members.
- Ask your PI and check current terms using the official links in the guide.

Neither service is required for this pre-work. Do not buy a plan for it.

Create a GitHub account (optional)

- Reuse your existing personal account, or open github.com/signup.
- Follow the signup prompts; choose a username and verify your email.
- Use a free account; no paid plan is needed for this optional setup.
- Enable two-factor authentication and keep recovery codes private.
- Check you can sign in. Education verification is a separate step.

GitHub: creating a personal account (checked November 9, 2026)

GitHub Copilot in VS Code (optional)

- Update VS Code. Open Extensions and search for GitHub Copilot.
- Use the official GitHub extension (GitHub.copilot-chat); install if needed.
- Recent VS Code: Copilot icon > Use AI Features also starts setup.
- Sign in with the account that has Copilot access; Free has usage limits.
- In a scratch folder, ask: Explain this hello.py script. Review the answer.

Official VS Code Copilot setup and sign-in (checked November 9, 2026)

GitHub Copilot desktop app (optional)

- Install Git first. Download from github.com/github/app for your OS and CPU.
- Run the platform installer, open the app, and sign in to GitHub.
- Use existing Copilot access, Free, or verified Student access if eligible.
- Open a disposable local folder; start an Interactive agent session.
- Ask for an explanation first; review changes and commands before approval.

This is the GitHub Copilot app, not GitHub Desktop or Microsoft Copilot.

GitHub Copilot app: installation and first session (checked November 9, 2026)

Install Freebuff (optional)

- Before installation: use a separate, isolated OS account/environment; otherwise skip.
- Download the official desktop installer for your OS and CPU at freebuff.com.
- Read current terms/privacy; install only in that isolated environment.
- Open the app and sign in; do not connect research drives or private repositories.
- Ad-funded optional tool: verify availability. You do not need every tool.

Freebuff official desktop downloads (checked November 9, 2026)

Use Freebuff safely

- Create hello.py with print("Hello, bioinformatics!") in a disposable practice folder.
- Use only public/synthetic files and keep an unchanged backup.
- Agents may access files beyond a project folder; it is not a sandbox.
- Try: Explain hello.py. Do not edit files or run commands.
- Review any proposed edits and commands; check the outputs yourself.

Read data handling and advertising terms. A local app is not necessarily local-only.

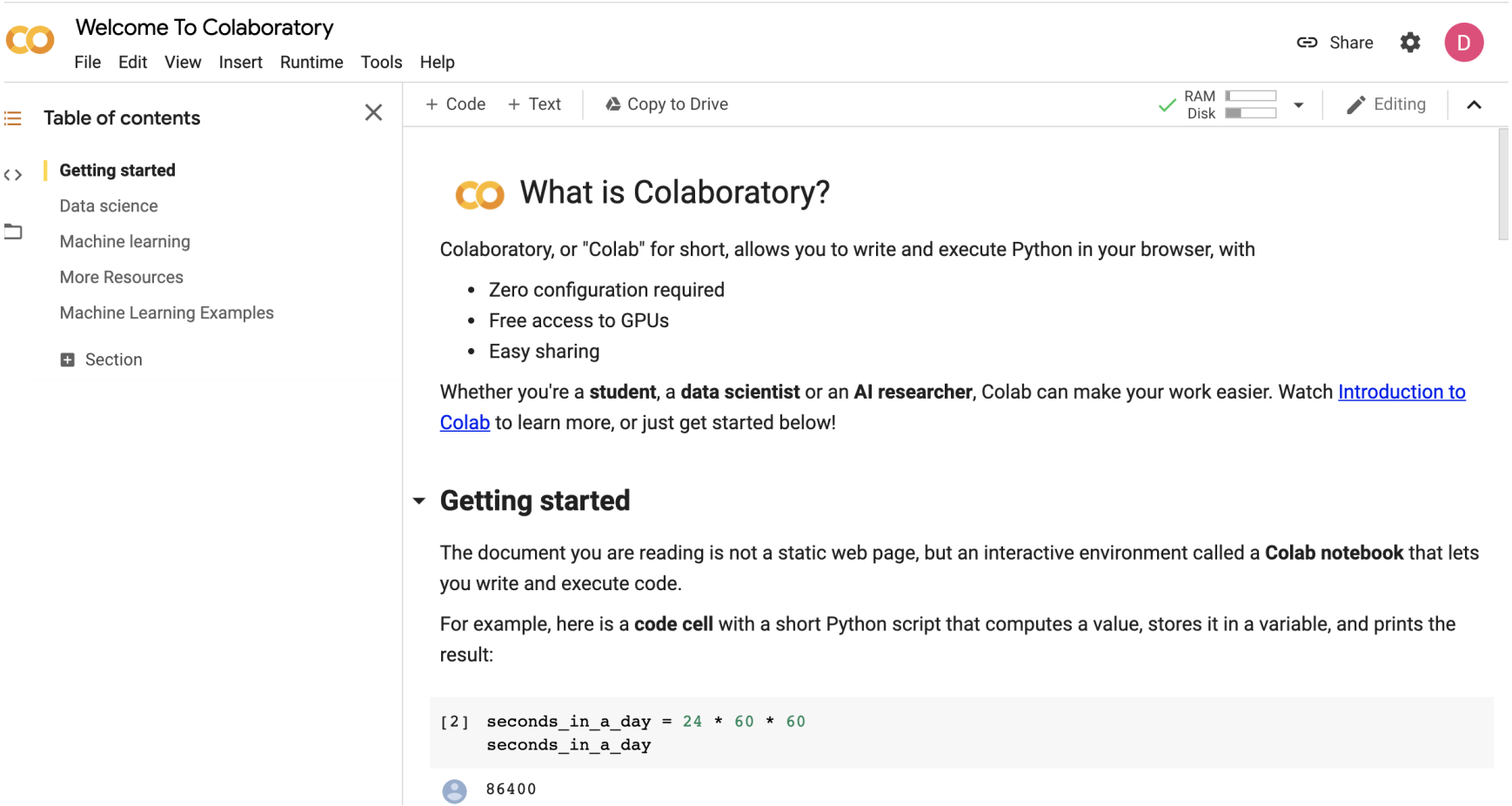
Freebuff privacy policy (effective November 12, 2026) (checked November 9, 2026)

Bring your own data (optional)

- Workshop example data will be available; bringing your own data is optional.
- Bring a small public or synthetic sample you are allowed to use, plus a question.
- Add a README / data dictionary: columns, units, provenance, and expected output.
- Do not upload patient-level, confidential, restricted, or unpublished data.
- Ask an instructor before using it; a private repository is not approval for AI use.

If your real data are sensitive, bring the schema or a synthetic example instead.

Inside a Colab notebook



Historical MLE 0426 Module 1 example. The following seven slides are original workshop slides; current Colab screens and features may differ.

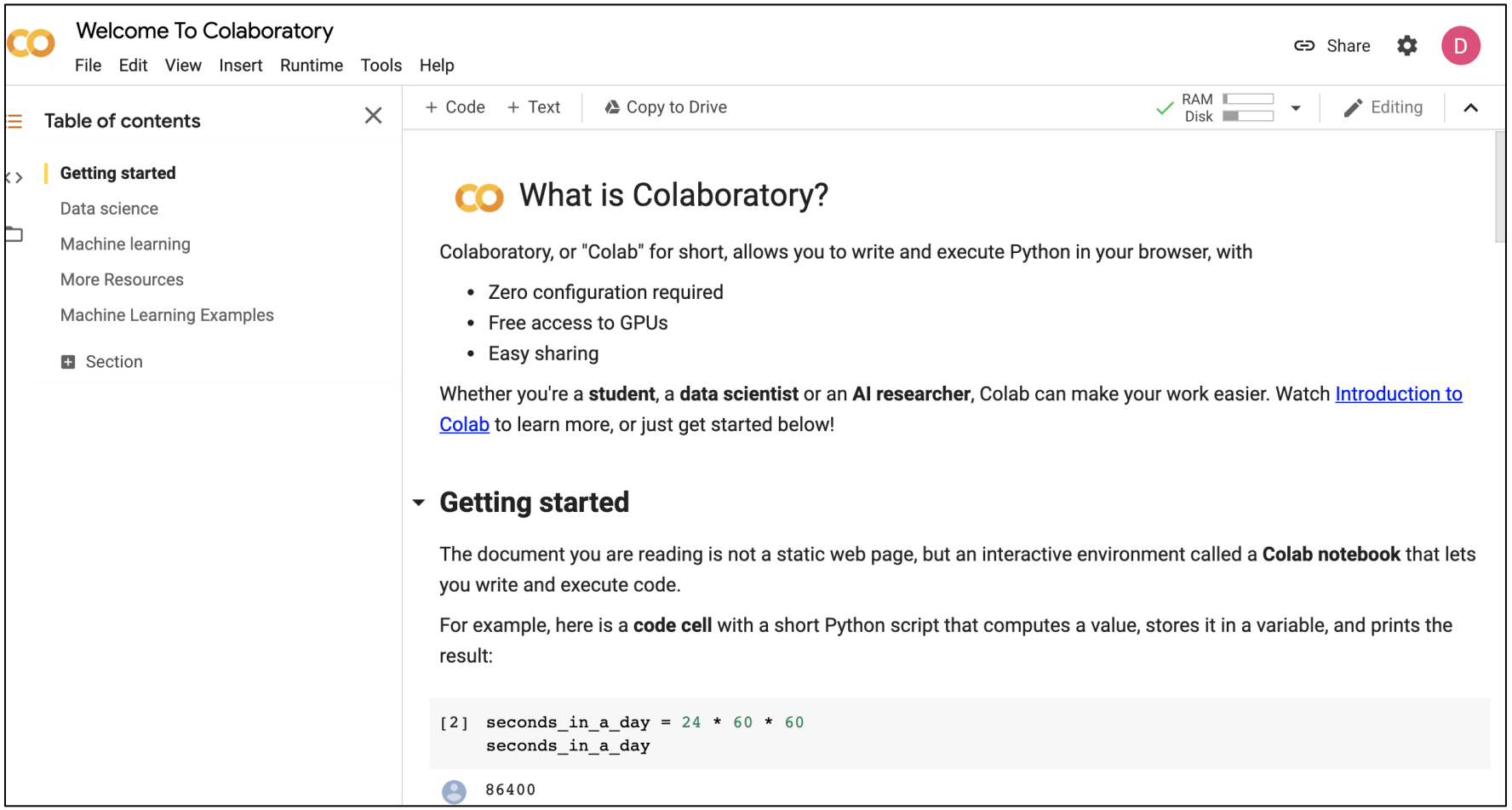
Google Colab

- Colab is a Python development environment that runs in the browser using Google Cloud
- Need to have a google account/google drive (see next slide)
- No need to install or run Python locally or deal with local OS issues
- Allows you to access, import and edit previously written Python code
- Allows you to share, save and download code
- Allows easy access to NumPy
- Allows easy access to GPUs, ML code and TensorFlow for ML applications

Google Colaboratory

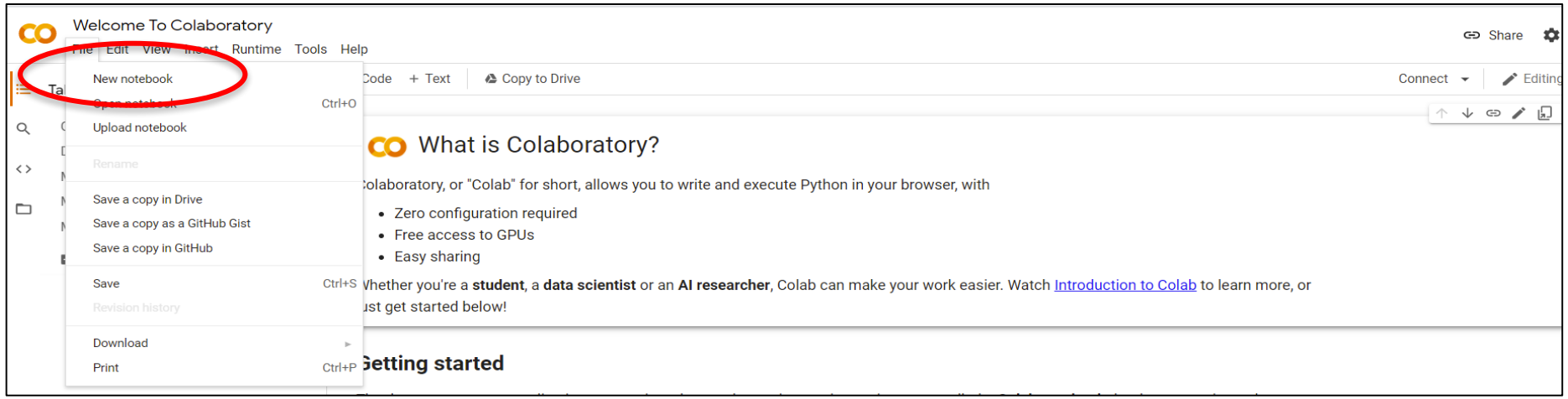
- Access by typing “Colab” in Google or go to https://colab.research.google.com/
- Click on “Welcome to Colaboratory” to launch
- Click on the “Introduction to Colab” video to learn more (3 minutes)
- Colab home page has File/Edit/View/Insert functions on upper left of page
- Runs like a Jupyter notebook
- Colab notebooks allow you to combine executable code, rich text, images, HTML, etc.

Colab Main Page



How to Open a New Notebook

- **Step 2: Under File, click New Notebook**
- **A new notebook should open up as shown below**





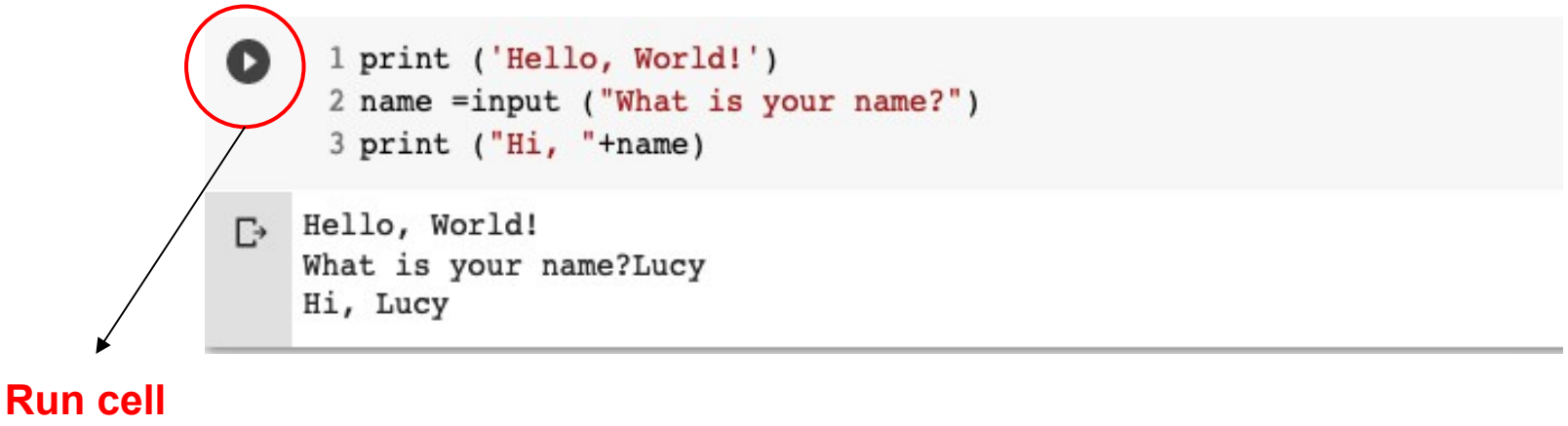
Hello World

- Let's write a program in Colab and call it "Hello World"
- Type that name in the circle and press Enter
- The notebook will acquire the name you have given now



Writing and Executing the Code

- Enter the trivial Python code and run it
- We will show how the simple code “Hello World” can be written
- Write this few lines of code in code cell
- Press run cell
- The output are the results



Runtime – Run All

- To save time, you can also click ‘Runtime’ then select ‘Run all’ to run all the cells

